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Abbreviations

CMOS complementary metal-oxide-semiconductor. 2, 24

CNT carbon nanotube. 1, 14, 15, 16, 22, 30, 32

CNTFET carbon nanotube field-effect transistor. 15, 18, 20

CVD chemical vapour deposition. 2

DFT Density Functional Theory. 21

DIBL drain induced barrier lowering. 18

EOT effective oxide thickness. 19, 20

FET field-effect transistor. 8, 15, 29

GNR graphene nanoribbon. 1, 22, 27, 28, 30, 31

GNRFET graphene nanoribbon field-effect transistor. 28

MOSFET metal–oxide–semiconductor field-effect transistor. 11, 12, 13, 19

MW multi-walled. 14, 32

NEGF non-equilibrium Green function. 21

QD quantum dot. 32, 34, 35

SB Schottky barrier. 15

SS Sub-threshold swing. 18, 19, 20

SW single-walled. 14

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Introduction

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In this chapter, I seek to give the motivation and context for this thesis: the investigation of molecular synthesised graphene nanoribbons (GNRs) for use in nanoelectronics. To this end, I will provide a brief introduction of the major all-carbon alternatives already investigated. The chapter is structured as to present first the solid-state crystal properties, and how these lead to unique and different dispersion relations across graphene, carbon nanotubes (CNTs), and GNR. Secondly, the most relevant applications and limitations are outlined.

1.1 Motivation

The widespread shortage of semiconductors begun in the late 2020 has called both major chipmakers and the scientific community to think more deeply and actively to beyond-silicon replacements. For largely over three decades, the main research-driven factor in the transistor industry has been the complementary metal-oxide-semiconductor (CMOS) density scaling; now it is leaving the floor to pursuit short-channel-effects robust materials and which are capable of being integrated in silicon production pipelines. According to Bloomberg, the basic costs of building a semiconductor factory are around 15 with profit requirements in the order of 3 to sustain market competition and avoid becoming obsolete [1]. These figures are also supported by an analysis done last year by the Boston Consulting Group, which places the investments in R&D (22% of annual semiconductors sales) and capital expenditure (26%) as the highest across all industries.

graphene in silicon pipelines

The large market position and capitalisaiton of chip makers industry have induced the proposals of integrating graphene in the semiconductor manufacturing cycle. Typically these are split between front- and back-end lines: while the former use temperature as high as 1000 K for activating the dopant process, back-end lines require milder temperatures (approximately 450 K), thus offering possible graphene integration solutions. Chemical vapour deposition (CVD) through plasma enhanced techinques has permitted the direct growth of graphene on silicon. The amorphous nature of Si and the presence of defects, however, are major sources of poor perfomance due to defects induced in the graphene layer. This circumstances could be mitigated by sputtering Cu particles onto the surface on top of the substrate, followed by graphene growth and metal removal. The last step can be problematic, as removing Cu catalyst requires etchants and high temperatures not compatible with back-end lines.

Decoupling the growth process from the final substrate offers a valid alternative: graphene is CVD-grown of Cu or Pt substrate and then transer graphene onto the target surface. Whilst suitable mobility values in the order of 10,000 $cm^2V^{-1}s^{-1}$ have been proved, this process has two major hurdles. Firstly, wafer-scale transferring techniques use polymers to handle graphene. Generally these are all high-molecular-weight polymers, such as PMMA, which are also flexible, to limit the surface tensions on graphene that could cause ripples and folds. Secondly, delaminating the metal substrate leaves contaminants in the graphene layer, causing unwanted doping effects.

1.2 Graphene

1.2.1 crystal structure

Graphene has a hexagonal structure where the carbon atoms are hybridised sp^2, as detailed by the Pauling theory on chemical bonds. Based on the experimental evidence collected in the last 20 years, the model that seems to best describe the proprierties of graphene is the band-structure Wallace model, which dates back in 1947, and describes graphene as a semimetal with a linear dispersion of the electronic excitations. In the International Table of Crystallography, graphene is included in the symmorphic space group #191, with symmetry D_6^1h ($P6/mmm$). The triangular lattice is obtained by considering two atoms per unit cell having the respective lattice vectors:

The carbon atoms in graphene form a two-dimensional hexagonal crystal with two atoms per unit cell, as shown in figure.

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}),\ \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \tag{1.1}$$

with $a \approx 1.42$ Å be the carbon-carbon bond length. Each atom belonging to one sublattice, called sublattice A, has three first neighbours belonging to the other sublattice, called sublattice B. With respect to one atom chosen from sublattice A as origin, the position of these first three neighbours in real space are given by:

$$\boldsymbol{\delta}_1 = \frac{a}{2}(1, \sqrt{3}),\ \boldsymbol{\delta}_2 = \frac{a}{2}(1, -\sqrt{3}),\ \boldsymbol{\delta}_3 = -a(1, 0), \tag{1.2}$$

The reciprocal-lattice vectors are then given

$$\mathbf{b}_1 = \frac{2\pi}{3a}(1, \sqrt{3}),\ \mathbf{b}_2 = \frac{2\pi}{3a}(1, -\sqrt{3}), \tag{1.3}$$

In order to define the vertices of the Brillouin zone one has to apply the Bragg condition:

$$2\mathbf{k} \cdot \mathbf{G} = 0 \tag{1.4}$$

where

$$\mathbf{G} = n\mathbf{b}_1 + m\mathbf{b}_2, \tag{1.5}$$

The first amongst these are:

$$\begin{aligned}
&n = \pm 1, && m = 0, && \mathbf{G} = \pm\mathbf{b}_1,\\
&n = 0, && m = \pm 1, && \mathbf{G} = \pm\mathbf{b}_2,\\
&n = \pm 1, && m = \pm 1, && \mathbf{G} = \pm(\mathbf{b}_1 + \mathbf{b}_2),
\end{aligned}$$

and all these vectors have the same modulus $|\mathbf{G}| = \frac{4\pi}{3a}$. By applying Bragg's condition in the first of the three cases we get:

$$\sqrt{3}k_x + k_y = \mp\, \frac{4\pi}{3a}, \tag{1.6}$$

which gets satisfied by the points:

$$K \equiv \mp\, \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, 1\right) \tag{1.7}$$

For the second case one gets:

$$\mp\, (\sqrt{3}k_x - k_y) = \frac{4\pi}{3a}, \tag{1.8}$$

which gives the points:

$$K' \equiv \pm\, \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, -1\right) \tag{1.9}$$

The third and last case gives the point:

$$M \equiv \pm\Big(0, \pm\,\frac{2\pi}{3a}\Big) \tag{1.10}$$

1.2.2 band dispersion

The electronic structure of graphene can be model with a tigh-binding hamiltonian. Following the second quantization formalism ($\hbar = 1$ units), the hamiltonian can be written as:

$$\mathcal{H}_{\mathrm{TB}} = -t \sum_{\langle ij\rangle,\ \sigma} \Big(a^{\dagger}_{\sigma,i} b_{\sigma,j} + \mathrm{H.c.}\Big) - t' \sum_{\langle\langle ij\rangle\rangle,\sigma} \Big(a^{\dagger}_{\sigma,i} a_{\sigma,j} + b^{\dagger}_{\sigma,i} b_{\sigma,j} + \mathrm{H.c.}\Big), \tag{1.11}$$

The notation $\langle ij\rangle$ means that the sum runs over pairs of electrons which are nearest neighbour; similarly, the notation $\langle\langle ij\rangle\rangle$ specify a summation over the next nearest neighbours; the operators $a^{\dagger}_{i,\sigma}$ ($a_{i,\sigma}$) and $b^{\dagger}_{i,\sigma}$ ($b_{i,\sigma}$) creates (annihilates) an electron with spin σ ($\sigma = \uparrow, \downarrow$) for atoms on sublattice A and B, respectively. The value of t which reproduces best experimental result is $t \approx 2.8$ eV and quantify the nearest-neighbour hopping energy; similarly, t' denotes the next nearest-neighbour hopping energy. The dispersion energy computed from this model reads:

$$E_{\pm}(\mathbf{k}) = \epsilon_p - t' f(\mathbf{k}) \pm t\sqrt{3 + f(\mathbf{k})}, \tag{1.12}$$

with $f(\mathbf{k})$ defined as:

$$f(\mathbf{k}) = 2\cos\left(\sqrt{3}k_y a\right) + 4\cos\left(\frac{\sqrt{3}}{2}k_y a\right)\cos\left(\frac{3}{2}k_x a\right). \tag{1.13}$$

Therefore, it is possible to identify an upper ($\pi*$) band associated with the plus sign and a lower (π) band described by the minus sign; these are symmetric with respect to the single-particle zero energy. The experimental estimate reported for next nearest-neighbour hopping energy is $t' \approx 0.1$eV thus the asymmetry introduced is small. It is worthwhile to notice that the function $3 + f(\mathbf{k})$

If the system is not doped, there is one electron per carbon atom, so two electrons per cell: it follows that the lower band is full and the upper band is empty the

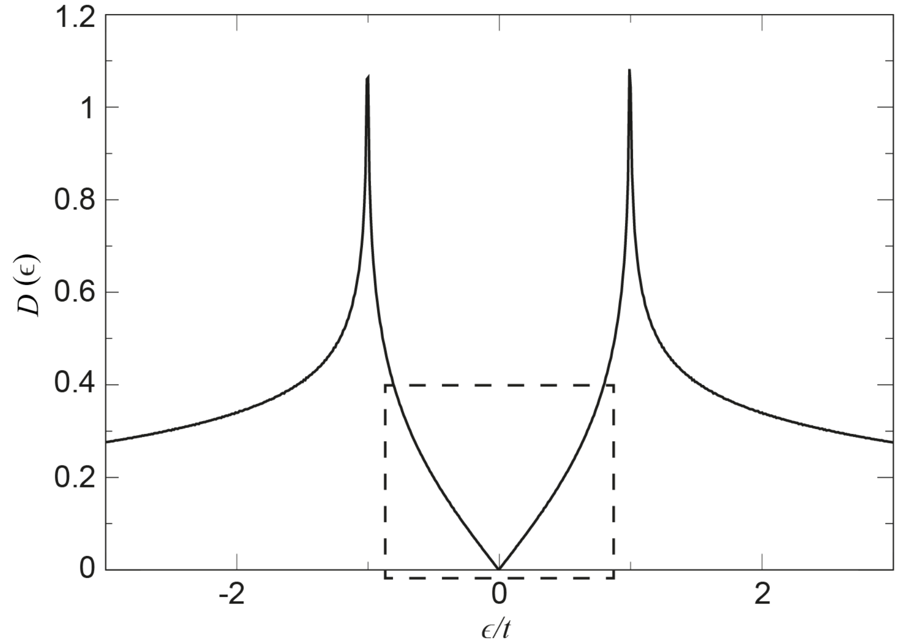

Figure 1.1: Graphene density of states plotted over a wide range of nearest neighbor hopping parameter t, showing pronounced van Hove singularities for $\epsilon \sim \pm t$ [2]. The dashed rectangle indicates the almost linear low energy DOS normally spanned in back-gated transport measurements.

upper one is empty. The system is therefore a semi-metal. The peculiarities of the dispersion in graphene become apparent when analysing their trend near the point **K** and **K'**. By taking the positive components for the **K** as define in Eq. 1.7 and expanding the Eq. 1.12 as $\mathbf{k} = \mathbf{K} + \mathbf{q}$, with $|(\mathbf{q})| \ll |(\mathbf{K})|$, one gets:

$$E_{\pm}(\mathbf{q}) = \epsilon_p \pm \frac{3at}{2}|\mathbf{q}| \tag{1.14}$$

which is linear in $\mathbf{q}$. For intrinsic graphene, at $T = 0\mathrm{K}$ $E_F = \epsilon_p$. The Fermi velocity is defined by the usuale relation $v_F = \frac{1}{\hbar}\left(\frac{\mathrm{d}E}{\mathrm{d}\mathbf{k}}\right)$, and in graphene $v_F \simeq 1 \times 10^6\ \mathrm{ms}^{-1} = 0.66\ \mathrm{eV} \times \mathrm{nm} \times \hbar^{-1}$.

A difference arises in the density of the states, when we limit ourselves to consider those near the Fermi energy. In the case of a two-dimensional electron gas, it is constant. On the contrary for graphene, the density of states per unite cell is:

$$\rho(E) = \frac{2A_{\mathrm{cell}}}{\pi}\frac{|E|}{v_F^2} \tag{1.15}$$

If the graphene is charge-neautral, the DOS has two Van Hove singularities at $\omega_{\mathrm{VH}} = \pm$ 1eV [3]. This behaviour is in stark constrat with the DOS of carbon nanotubes and clean graphene nanoribbons, as they show $1/\sqrt{E}$ singularities due to the 1D nature, which imposes a quantization perpendicular to the ribbon or tube longitudinal direction.

Using the Einstein-Smoluchowski relation within the semiclassical description, from the DOS it is possible to model the (semiclassical) conductivity as:

$$\sigma_{\mathrm{sc}} = e^2 \rho(E) v(E) \ell_{\mathrm{el}} \tag{1.16}$$

and from this one can calculate the charge mobility, which, at zero temperature, reads:

$$\mu(E) = \sigma_{\mathrm{sc}}(E)/en(E) \tag{1.17}$$

w is the semiclassical conductivity deduced from the Einstein formula, with $\rho(E)$ the DOS, (E) the charge density at energy E, ℓ_{el} the elastic mean free path, and e the elementary charge. The charge mobility is often used as a figure to assess the graphene strucure quality. Close to the Dirac points, the measured experimental conductivities are mostly found to range within $\approx 2 - 5e^2/h$, although the charge mobility can vary due to doping from the contacts and/or the substrate by almost one order of magnitude (Jiang et al. 2007;Oezyilmaz et al. 2007; Zhang et al. 2006).

1.2.3 Graphene Transistors

For transistor applications, a material needs a highly conductive on-state and an off-state where the source-drain current I_{sd} does not flow. For RFs applications, on the other hand, the full speed operation is obtained by operating in the (drain) current saturation regime. Both applications requires a sizeable band gap. Furthermore, the IC scaling trend is often juxtaposed to chip-level issues such as power densities (currently on the order of 1000 W/cm^2), heat generation, and chip temperatures reaching levels that will prevent reliable operations.

In CMOS logic, the charge carriers are accellerated by the electric field $\mathcal{E}_{\mathrm{sd}}$ between source and drain, thus undergoing five possible sources of scattering:

electron–electron scattering, electron–phonon (lattice vibrations), material interfaces, imperfections or impurity atoms. The first two are more fundamentals, while the last three are material-related. The most dominant mechanism through which charge carriers lose net energy is by scattering with phonons, consequently heating up the lattice via Joule heating. For RF applications, on the other hand, heat is produced by the operational frequency at which the carriers operate. Thus, two further requirements are needed: on one hand a good thermal conductivity, but also small heat transfer with the substrate. Clearly trade-offs are here necessary.

The 2020 International Roadmap for Devices and Systems - which has replaced since 2017 the International Roadmap for Semiconductors - dedicates ample attention to graphene in its report *Beyond CMOS*, in which technologies and semiconductor materials to complement and replace Si are considered. The race to scale the channel in a field-effect transistor (FET) still motivates a significant effort on introducing high-mobility channel materials, since the carrier mobility is an appropriate parameter to address carrier transport even in nanometer FETs [4]. In fact, most of the early excitements in using graphene as channel material in a standard FET technology were sparked by considering the outstanding mobility values reported since its discovery.

At 290 K, the intrinsic mobility μ_e^{in} of graphene is exceptionally high. The first data were the one obtained in 2004 on large-area graphene, showing an electron mobility $\mu_e^{\mathrm{in}} \approx 10,000$ cm^{2}V^{-1}s^{-1} [5]. Further studies have demonstrated mobilities spanning between $\mu^{\mathrm{in}} \sim 3000 - 12000$ cm^2 V^{-1}s^{-1} for carrier concentrations $n \approx 1 - 0.1 \times 10^{12}$ cm^{-2}, with low-field field-effect mobility† ($\mu = (1/C_{\mathrm{g}})(\partial\sigma/\partial V_{\mathrm{gs}})$) that is constant over a quite wide range of n (yielding in this case $\mu_h \approx 1900$ cm^2 V^{-1} s^{-1} and $\mu_e \approx 1800$ cm^2 V^{-1} s^{-1} for the low n region) [6].

The mobility is limited by different factors such as scattering by charged impurities or defects and electron-phonon scattering. In very clean and suspended samples (where the underling SiO_2 substrate has been etched away) the mean free

†This differs from the Drude ($\mu_{\mathrm{Drude}} = \sigma/ne$) and Hall ($\mu_{\mathrm{Hall}} = -\sigma_e R_H$) mobilities.

path l increases by almost two order of magnitude, becoming comparabe to the device size, an indication that the ballistic transport regime is reached

Later in 2008 [7] and 2013 [8], higher mobility values were reported: $\approx 120,000$ cm^{2}V^{-1}s^{-1} at 240 K and $\approx 1,000,000$ cm^{2}V^{-1}s^{-1} at 4 K. The reason for these higher values was attributed to the use of suspended graphene rather onto substrate, where surface effects inevitably affect the carrier dynamics. At the same time, results ($\approx 120,000$ cm^{2}V^{-1}s^{-1}) obtained at room temperature with graphene onto a single-layer of hexagonal boron nitride (hBN) [9] and sandwiched between two hBN layer [8] have suggested that this flat and inert materials can be act as a excellent passivation film for graphene devices.

One of the figure of merit often taken into account for addressing logic applications for a materials is the ratio between the on-state current and the off-state current. The so called on-off ratio $I_{\rm on}/I_{\rm off}$ has to be in the range 10^4–10^7. While the $I_{\rm on}$ is not directly related to the band gap E_g, it is possible to link the value of $I_{\rm off}$ with the E_g.

In an metal–semiconductor junction (M/S), the current is built by the majority of the carriers, these being electrons or holes for n-type or p-type semiconductors. Most of the graphene–semiconductor junctions formed during experiments have typical dopant concentration $N \leq 10^{17}$ cm^{-3}. The $I-V$ characteristics of M/S junctions with such dopant concentrations are well reproduced by the thermoionic emission theory. If we then exclude other possible conduction mechanisms (standard and thermoionic field emission, generation/recombination, diffusion of electron, injection of holes and leakage current), the $I_{\rm off}$ becomes proportional to the barrier the electrons have to overcome from the semiconductor to the metal, namely

$$I_{\rm off} \propto \exp\{(e\beta\phi_B)\}. \tag{1.18}$$

where e is the electron charge, ϕ_B is the Schottky barrier height, and $\beta = k_B T$. If we assume that the work function of graphene and the metal are the same, then $\phi_B = E_g/2$, where E_g is the band gap energy originating from the difference of the work function of grahene and the metal; thus, the on–off ratio can be estimated from the useful proportionality [10]:

	Direct Gap			Indirect Gap		
	InP	Si	Ge	GaAs	InAs	Graphene
a_{lattice} (Å)	5.87	5.43	5.65	5.65	6.06	2.46
E_g (eV)	1.35	1.12	0.66	1.42	0.36	0
μ ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$) at 300 K	2800	1400	3900	4600	16,000	200,000
v_{sat} (10^7 cm s^{-1})	2.2	1	0.6	2.2	4.0	>5
DOS (m^*/m_0)	0.077	1.08	0.56	0.067	0.023	0
ϵ_r	12.4	11.9	16.0	13.1	14.6	2.4
κ (W m^{-1} K^{-1})	68	150	60.2	46	27	5000

Table 1.1: Differences between the main semiconductors used in modern digital electronics and graphene.

$$\frac{I_{\text{on}}}{I_{\text{off}}} \propto \exp\left\{\frac{E_g}{2k_BT}\right\}, \tag{1.19}$$

where the thermal energy k_BT and the bandgap are both expressed in meV. While for high frequency applications the on–off ratio is of minor importance, several references in literature report suitable values for CMOS bandgap should fall in a range between 300 and 500 meV [11, 12].

Hysteresis between the forward and reverse sweeps have been reported depending on the sweeping rate and range [13], and attributed mainly to charge trapping in silanol groups (Si–OH), mediated and assisted by surface-bound H_2O molecules [14]. Despite the usefulness for its proof-of-concept purposes, back-gate devices suffered from parasitic capacitances, making impossible further integration with other components. Alternatives geometries have since then been explored, in order to increase the carriers mobility, to decrease contact resistances and parasitic capacitances, and to avoid Schottky barriers formation between the graphene and the semiconductor. Top-gate devices have been the first to chronologically appear in the literature. Back then in 2007, Lemme and co workeres used a stack of SiO_2 (20 nm), Ti (10 nm), and Au (100 nm) as top gate for modulating the current inside a large-area graphene channel transistor obtained by the exfoliation method [15]. Whilst the gate characteristics exhibit a similar trend of the characteristics (the top gate reduces the back-gate current modulation, lowering the drain current by one order of magnitude) and the hysteresis, weak current saturation at high drain-source bias have been reported [16]. This effect seem due to a partial charge pinning at the contacts at high back-gate voltages [17] and carrier drift velocity eventually saturating due to optical-phonon scattering [18]. Top-gated graphene metal–oxide–semiconductor field-effect transistors (MOSFETs) with CVD and epitaxial graphene have been also made [19, 20]; SiO_2, Al_2O_3, and HfO_2 have been used as top-gate dielectric.

However, despite the improvements obtained with a top gate, most graphene MOSFET with large-area graphene channels show on–off switching ratios of less than 10 at room temperature [11], much too little for complex logic circuits.

In 2012, Mehr *et al.*. proposed a vertical arrangement of emitter, base, and collector similar to the original vacuum triode, where graphene plays the base role and electrons travel vertically rather than transversely. When the device is in the ON state, the electrons tunnel with a Fowler-Nordheim tunnelling mechanism:

$$I \propto \left(\frac{V_{\mathrm{sd}}}{d}\right)^2 \exp\left\{-\frac{\alpha\ d}{V_{sd}}\right\}, \tag{1.20}$$

through the emitter–base layer (of thickness d); ON/OFF ratios of the order of 10^4 have been demonstrated using graphene between two h-BN layers [21] and using WeS_2–graphene [22] heterostructures.

Solution methods have also been explored, in which graphene is dropcasted on top of a semiconducting wafer, and then spin-coated [23]. The main difficulties arise when attempting to quantify the structure and the quality of the deposited graphene, and not many methods can guarantee a good dispersion in organic solvents [24]. Simpler ways to obtain graphene oxide in solution have been reported [25], but once deposited, graphene oxide needs to be reduced for obtaining pure graphene –

low mobility values (up to, at best, 1000 $cm^2\,V^{-1}\,s^{-1}$) caused by defects during the fabrication steps have discouraged further investigations along this direction.

Yet in spite of the very different dc behavior compared to the most used semiconductors in digital electroncis, graphene FETs have reported good performance in terms of high-frequency applications. This comparison is made by measuring each system's frequency response, its output signal strength when it is given a particular input signal: features such as ballistic transport, linear current-voltage characteristics, high transconductance, and very high ($> 10^8$ $A\,cm^{-2}$) sustainable currents are all ideal features for analog/RF applications [18, 26].

The primary means of estimating frequency response is through the evaluation of two transistor charactersitic frequencies, namely cutoff f_T and maximum of oscillations f_{max}. The former indicates how quickly a signal can travel from the gate to the drain of the device, and it is usually determined when the small-signal current gain drops to unity, while the latter is measured when the power gain decreseas to unity [11]. The frequency response depends inversely on gate length and proportionally to the transconductance, which is loosely proportional to mobility. Thus, reducing the gate length and using higher mobility materials are valid approaches to increase the device operational speed [27]. In this context, many research groups have successfully demonstrated RF capabilities with gapless graphene channels MOSFETs.

The first device with cutoff frequency f_T in the GHz range was reported in late 2008, and was farbicated by exfoliated graphene deposited on top of high-resistivity Si substrate, with 30-nm-thick top-gate HfO_2 dielectric and 500 nm of gate length [26]. Despite a measured $f_T \approx 14.7$ GHz, it was argued that large contact resistances and gate length were the main responsible of poor perfomance $f_{max} < 1$ GHz.

The following year, Pd/Au contacts and 12-nm-thick Al_2O_3 were employed in a 350-nm-gate [28]. While the cutoff frequency improved up to $f_T \approx 50$ GHz, little gains in terms of f_{max} were obtained of only 2 GHz. In general, limited performance in f_{max} represents a major issue as for most RF applications f_T is less important than f_{max}.

It was only after two years that an increase of one order of magnitude both in the cutoff and the maximum frequency of oscillation was demosntrated [20]. The device was realised with epitaxially-grown on silicon carbide (SiC) graphene MOSFET having 240 nm of gate length, with Ti/Pd/Au for contacts and 10-nm-thick HfO_2 of insulating layer on top gates. Highest frequency as well as $f_T \approx 100$ GHz and $f_{max} \approx 14$ GHz were measured, exceeding any device then present. The authors also argued that since f_{max} is strongly correlated to device layout and parasitic resistances originating from the metallic contacts, viable ways to improve this parameter would be a partial passivation of the metal contacts and a multi-finger gate layout

These perfomance were soon followed later that year by a 144-nm-length gate device with $f_T \approx 300$ GHz using mechanically exfoliated graphene and Co_2Si–Al_2O_3 nanowire as self-aligned multi top-gates [29]. This result was also corrobated by comparing it with the projected value using the relation that links dc transconductance and gate capacitance with the cutoff frequency when total input parasitic capacitance is negligible, namely $f_T = g_m/(2\pi C_g)$ [30]. The authors did not report the power $G_{max}^{1/2}$ but only the current $|h_{21}|$ gain, thus hampering an evaluation of f_{max}, but in 2012 a similar proposal was advanced by Cheng *et al.*, where Au/Ti/Al_2O_3 gate stacks were employed in self-aligned transistors [31]. Despite the improvements reported ($f_T \approx 427$ GHz for a channel length of 67 nm) respect to CVD graphene transistors with comparable channel length, f_{max} decresead down to 8 GHz when the length was shrunk down to 46 nm, showing the key role played by gate and parasitic resistances at high frequency.

To further improve the power gain, works on the saturation behavior of short channel graphene transistor were also conducted by IBM in 2012, by using CVD graphene grown on a Cu foil and then transferred on diamond-like and SiC substrates [32]. These are appealing because high-quality wafers are available, thus making a scale up of graphene arrays relatively straightforward. Since the work function of graphene is $\phi^{g} = 4.5$ eV and $\phi^{4H\text{-}SiC} = 3.7$ eV, this means that $\Delta\phi^{g\text{–}4H\text{-}SiC} = 0.8$ eV, thus n-doping graphene [33]. Very cleverly, the authors then used plasma-enhanced CVD silicon nitride as an intrisic p-type gate dielectric, counterbalancing the n-doped graphene channels. Encouraing performance at high speed were not only measured in good cutoff frequencies ($f_T \approx 350$ GHz), but also in a factor of 2 improvement of f_{max} (≈ 40 GHz). To the best of my knowledge, the current graphene transistor power gain record of about 70 GHz was obtained in 2013 by using (000-1) carbon-terminated surface of SiC instead of the (0001) Si one, indicating the contact resistance R_c between Pd/Au contacts and graphene as key responsible for the high f_{max}.

While these results have aligned graphene MOSFET with the main solid-state competitors – such as GaAS pHEMT, InP HEMT and Si MOSFET – in terms of f_T, the perfomance is terms of f_{max} are off by about one [34, 35] or two orders of magnitude [36]. In literature, one of the often cited fundamental reason for this behaviour is the poor drain saturation current in graphene, which ultimately depends on the missing bandgap of the channel. Graphene-derived materials such as nanoribbons and carbon nanotubes MOSFET could then, given their intrisic semiconducting nature, represent a possible solution to this problem.

1.3 Carbon-based Field Effect Transistor

1.3.1 crystal structure

This crystalline form of carbon can be considered as a rolled up sheet of graphene and comes in two major flavours: single-walled (SW) and multi-walled (MW). CNTs display an electronic band gap which depends on the chirality and the diameter of the nanotube [37]. Having defined two unit vectors in the real space, $\mathbf{a_1}$ and $\mathbf{a_2}$, a CNT can be described by the linear combination of these with integers coefficients (m, n). Based on the values of m and n, CNTs can be metallic , quasi-metallic ($m-n$ is a multiple of 3, with $m \cdot n \neq 0$ and $m \neq n$), and semiconducting for the majority of the remaining cases – exceptions due to the curvature (spin–orbit couplings) in small-diameter tubes are also known [38]. Thus, a quantization of the electronic states in the circumferential direction is resulted when the nanotube diameter d_{CNT} is between 0.8 and 2.5 nm. Each of these substates is called subband and are determined by the periodic boundary condition. Having defined a circumferential wave vector $\mathbf{k_c}$ such that $\mathbf{k_c} \cdot \mathbf{c} = k_c|c| = k_c \pi d_{\mathrm{CNT}} = 2\pi\nu$, for ν integer, it is possible from this relation to draw parallel lines in the first Brillouin zone of graphene. If the subband intersect one of the $\mathbf{K}$ points, than the resulting nanotube will have no energy gap and thus exhibit metallic properties (this condition is realised by having $m = n$ for the coefficients of the chiral vector). If none of the subbands pass through $\mathbf{K}$ then the resulting CNT will be semiconducting, having a E_g proportional to the smallest distance between a $\mathbf{K}$ point and the subband lines:

$$E_g = 2\left(\frac{\partial E}{\partial k}\right)\frac{k_c}{3} = \frac{2}{3}\hbar v_F \frac{2\pi}{|c|} = \frac{4}{3}\hbar v_F \frac{1}{d_{\mathrm{CNT}}} \approx \frac{0.7}{d_{\mathrm{CNT}}} \tag{1.21}$$

where $v_F = 8 \times 10^5\ \mathrm{m\,s^{-1}}$ is the Fermi velocity, E_g is expressed in eV, and d_{CNT} is expressed in nm.

1.3.2 Carbon Nanotube Field-Effect Transistors

General Operation

The modulation of a CNT channel by a gate-induced field was realised in the 1998 by the group of Cees Dekkar. These early FETs were fabricated by using doped silicon substrate as a gate to modulate the energy bands, separated from the nanotube by thermally-grown SiO_2 oxide acting as dielectric. For contacting the nanotube, metal conctats were employed.

Since then, a great variaty of carbon nanotube field-effect transistor (CNTFET) structures and geometries have been demonstrated, and can be essentially classified in devices with or without a spacer. where the spacer length L_{spr} is the distance between the gated channel and the source/drain metal conctacts. Given that the nanotubes are intrinsically semiconducting, in spacer-less devices charge carriers are not induced in the channel by the gate-field but are injected from the contacts: the role of the gate is to lower the thermal barrier in the channel, thus allowing a tunnelling through the Schottky barrier (SB) at the source. In this scenario, a critical role for the CNTFET performance is played by the metal work function ϕ^{M}.

CNTFETs with spacers, on the other hand, are less limited by the injection at the junction source–CNT, as they are regulated by the modulating effect played by in the channel as in conventional. Since the carrier barrier injection at the drain is large, device with spacers have been proved to be more resistant to ambipolar conduction. In Table 1.2 an overview of gated-spacer (top and bottom) devices is reported.

Contacts

One of the principal issues when study the interface metal–CNT is the role played by carbon and metal atoms at the interface. Solid-state semiconductors have dagling bonds at the surface, leading to Fermi level pinning, meaning that the variation of the band bending is independent of the work function of the metal, even when this is large. Since the ideal structure of CNTs has no open bonds, no pinning of the metal Fermi level to mid-gap states occurs, thus legitimating the study of different work functions for maximising performance and tuning polarity. Many metals have been studied for contacting CNTs. Interestingly, it has been demonstrated that Pd ($\phi^{\mathrm{Pd}} = 5.22 - -5.60$) builds an almost ideal p-channel contact to the valence band of CNTs. Despite having a larger workfunction ($\phi^{\mathrm{Pt}} \approx 5.65$ eV) than Pd, Pt provides very poor contacts due to the physical wetting of the metal on the nanotube.

Device type	$I_{\rm on}/I_{\rm off}$	SS (mV/dec)	$g_{\rm mt}$ (µS cm^{-1})	$L_{\rm ch}$ (nm)	f_T [$f_{\rm max}$] (GHz)	Ref.
	10^4	94	~ 40	9	—	[39]
	10^5	~ 85	40	15	—	[40]
	10^5	~ 85	20	20	—	[40]
	10^5	60	20	200	—	[41]
Bottom gate	10^5	~ 63	—	200	—	[42]
	10^4	80	50	240	—	[43]
	—	—	143	800	[1]	[44]
	10^4	—	—	1,500	1.5 – 3.0	[45]
	—	70	$12^\dagger$	2,000*	—	[46]
	10^8	—	—	$< 10,000^*$	—	[47]
	—	—	- 500	—	30	[48]
	10^6	—	$20^\dagger$	80	—	[49]
Top gate	2	—	~ 50	100	25	[50]
	50	—	310	100	~ 70 [80]	[51]
	—	—	0.3	210	—	[52]
	10^5	—	33.5	85,000	—	[53]

Table 1.2: Figures of merit for a few CNFET reported in literature. S: subthreshold swing; $g_{\rm m}$: intrinsic transconductances; L_{ch}: channel length; μ: carriers mobility

†Normalized by the tube width.
^{1} Referred to the gated length of a nanotube.
^{2} For the n-type transistor, whereas 10 µS were measured for the p-type.

Device type	$I_{\rm on}$ (A)	$I_{\rm off}$ (A)	$I_{\rm on}/I_{\rm off}$	$g_{\rm mt}$ (µS)	SS (mV/dec)
Conventional	$1.41{*}10^{-5}$	$4.06{*}10^{-10}$	$3.4{*}10^{4}$	29	99.6
Partial S/D-doped	$1.50{*}10^{-5}$	$2.33{*}10^{-8}$	$6.4{*}10^{2}$	63.6	93.1
Schottky-Barrier	$3.26{*}10^{-6}$	$2.31{*}10^{-8}$	$1.41{*}10^{2}$	19.6	109.8
Tunnel	$5.53{*}10^{-6}$	$1.16{*}10^{-8}$	$4.76{*}10^{2}$	13	107.9

Table 1.3: Comparative simulation for the four major type of CNFETs as reported in [54]. These results are obtained by setting $V_{\rm sd} = 0.5$ V.

n-Type CNTFETs

In a n-type CNTFETs the injection of electrons is favoured over the holes. Given that $\phi^{\mathrm{CNT}} \approx 4.7$ eV and with an average $E_g \approx 0.6$ eV, it is possible to configure n-type channel with nanotube by either doping the spacer region or by using a metal with lower work function. One of the main drawbakcs of doping is the challenge to keep it under control: instability in air, low consistency, variability to temperature variations, are amongst the most frequent sceanrios. The other issue in working with low work function metals as contacts is the high tendency to oxidize. A simple and effective workaround to this is to use high work function metals to cap the contact metals, passivating the contact surface tendency to destabilise in air. One example of high performance n-type device was obtained by using Y contacts, achieving an Sub-threshold swing (SS) of 73 mV/dec, mobility of 5100 $\mathrm{cm\,V^{-2}\,s^{-1}}$, and drain induced barrier lowering (DIBL) of 105 $\mathrm{mV\,V^{-1}}$ [55].

Scaling

While Si technology was for long time questioned whether it could work below 10-nm of channel length, recent impressive techological developements from the industry have proved not only its feasibility but also a widestream commercial application (currently modern laptop employ the 5-nm technology). Nevertheless, the interest in Si scalability has been always a crucial starting point for exploring different technologies and materials capable of overcoming short-channel effects (threshold voltage roll-off, off-state leakage current, channel length modulation are just a few of the problematics).

In carbon nanotubes transistors a crucial and important – but somehow overlooked difference – is the difference between the two critical lengths for scaling, which are the contact length (L_{c}) and the channel length (L_{ch}). Aggressive scaling for the channel length generally leads to lose control of the band bending through the gate: intuitively, by shrinking the channel length the electric field generated by the source contact will be felt more and more by the charge carriers. The argument here is that carbon nanotubes offer, however, a good enough platform for such scaling given by the ultrathin hollow structure (recall an effective wall thickness about $d_{\mathrm{CNT}} \approx 0.68$ Å). On the other hand, this could be a double edge-sword. The effetive carrier mass in nanotubes is $m^{*}_{\mathrm{CNT}} = (0.228 \pm 0.129)m_0$ [56], thus potentially leading to tunnelling current from source to drain: this effect can be appreciate by an increase in the SS and the OFF current.

One of the first few systematic study on channel length scaling effects was conducted by Franklin *et al.*. in 2010 through a series of measurements on nanotubes with relatively small diameters (1.1 nm $<$ $\mathrm{d_{CNT}}$ $<$ 1.3 nm) by using local bottom gates covered with 2 nm effective oxide thickness (EOT) of HfO_2 [57]. A consistent trend was observed when the channel length was scaled down more than two orders of magnitude (from 3 µm to 15 nm), both in the values of V_{th} and SS (-0.5 ± 0.2 V and 90 mV/dec, respectively). The authors reported also encouraging figures on the transconductance g_m.

In a standard MOSFET, the transconductance is the change in the drain current divided by the amount varied in the gate/source voltage V_{gs} when the drain/source voltage (V_{ds}) is kept constant. Typical values of g_m for a small-signal field effect transistor are between 1 and 30 $\mathrm{mS\,\mu m^{-1}}$. In a n-channel (nMOS) FET with threshold voltage V_{th}, the drain current at a certain V_{gs} in saturation mode (so when $V_{\mathrm{th}} > V_{\mathrm{gs}} - V_{\mathrm{sd}}$) is given by:

$$I_{\mathrm{sd}} = \frac{1}{2}\mu_n C_{\mathrm{ox}} \frac{W}{L}(V_{\mathrm{gs}} - V_{\mathrm{th}})^2, \tag{1.22}$$

where μ_{n} is the electron mobility, C_{ox} is the gate oxide capacitance, W and L are the gate width and length, respectively. It is possible to work out the value of g_m by using the Shichman–Hodges model and assuming that the threshold voltage V_{th} is a device constant, such that $V_{\mathrm{ov}} = V_{\mathrm{gs}} - V_{\mathrm{th}}$:

$$g_{\mathrm{m}} = \frac{\partial I_{\mathrm{sd}}}{\partial V_{\mathrm{gs}}} = \frac{2\ I_{\mathrm{sd}}}{V_{\mathrm{ov}}}, \tag{1.23}$$

Here V_{ov} is the so called *overdrive voltage* and has the physical meaning of being the voltage above the threshold voltage of the MOSFET. It is useful in the designing phase of a transistor and elegantly links the drain current with a voltage: in fact, by rearranging Eq. 1.22 and identify $k_n = \mu_n C_{\mathrm{ox}} \frac{W}{L}$ one gets:

$$I_{\mathrm{sd}} = \frac{1}{2} k_n V_{\mathrm{ov}}^2 \implies V_{\mathrm{ov}} = \sqrt{\frac{2 I_{\mathrm{sd}}}{k_n}} \tag{1.24}$$

The usefulness for this equation is that the relation expressed in terms of overdrive voltage rather thant threshold voltage holds both in the n and in the p case. Some reference numbers: V_{ov} is tipically set at about 70–200 mV for the 65 nm technology node (with $I_{\mathrm{sd}} = 1.13\ \mathrm{mA\,\mu m^{-1}}$ of width) with value of $g_{\mathrm{m}} \approx 11 - 32$ $\mathrm{mS\,\mu m^{-1}}$. In their work, Franklin *et al.* achieved transconductance peak up to 40 $\mathrm{\mu S\,\mu m^{-1}}$ for a CNTFET with 15 nm of channel length, corresponding to values of g_{m} between approximately 91 – 110 $\mathrm{\mu S\,\mu m^{-1}}$, thus meeting the requirements for being compatible with the Si technology.

The first CNTFET in which L_{ch} was scaled below 10 nm came two years later, again by Franklin *et al.* [39]. By using a CNT with $\mathrm{d_{CNT}} \approx 1.3$ nm $\pm$ 0.2 nm and $E_g \approx 0.62$ eV $\pm$ 0.1 eV contacted through local bottom gates, they manage to scale down the $L_{\mathrm{ch}} = 9$ nm The EOT was reduced to ≈ 0.65 nm to achieve a better electrostatic coupling with the nanotube.

The authors reported an increase on the minimum current as L_{ch} decreseas from 320 nm to 9 nm, attributed to a back-injection of the carriers into the conduction band from the drain, while the on-current approached the ballistic value for $L_{\mathrm{ch}} \leq 41$ nm [57]. Impressive switching behaviour, with SS about 94 mV/dec, was measured, thus far outperforming any silicon-based of that time. based on previous results [57], the authors claimed the possibility of lowering the SS and the off-current at minimal cost to Ion by reducing the diameter of the nanotubes.

Contact Length Scaling

MOSFET devices, when shrunk down to the nanoscale, need to consider not only L_{ch} but also the lenght of the nanotube with the source and drain, L_{c}. The first study to address the impact of contact scaling in CNTFETs revealed the L_{c} to be inversely dependent on the channel length. Initial theoretical attempts to describe this trend using dipole theories were started following the results obtained by the group of Hongjie Dai, where they showed it is possible to recover a perfect ohmic contact – thus accessing the ballistic transport regime – even with a semiconducting nanotube, which were at tha time notorious of suffering several Schottky barriers effects (tunnelling and thermoionic emission).

Further investigations using more sophisticated techiniques such as Density Functional Theory (DFT) coupled with the non-equilibrium Green function (NEGF) were employed to consider not only different contact geometries, but mostly to clarify the role in the chemical bonds formed between the metal to carbon structures, or in some combination of metal-/carbon-dependent interfacial charging properties. In particular, Cuniberti and coworkes addressed the chemisorption problem of carbon nanotubes with different electrodic surfaces. Chemisorption consists of the creation of a chemical bond between the atom deposited on a substrate (adatome) and its surface. The bonding leads to changes in the local density of states that produce specific alteration of the surface properties. They considered metal surfaces (such as Ni and Ti) inducing a chemical reaction with the carbon structure (chemisorption), elements such as Pd, Pt, Cr and Sc which interact weakly with the nanotube band structure leaving it unchanged, and surfaces which interact strongly through metallization and/or hybridisation of the π orbitals of the nanotube (Cu, Au, and Rh).

Low-ohmic contact scaling trends were found when using weakly and intermediate interacting metallic surfaces, while they envisioned optimal contact length scaling with metals such as Cu and Rh. While Rh contacts would be unaffordable at large scale due to manufacturing costs, it is well know that Cu present ill-defined native oxide issues when used under ambient conditions, causing scattering at their surfaces and grain boundaries, thus hampering good electrical contacts with external probes. Recently, it has been advanced graphene as interlayer between the metal contact and the nanotube: the idea would be to have a work function closer to the nanotube one in order to avoid the level pinning effects due to gap states and interface contaminations.

1.4 Graphene Nanoribbons

Amongst the wide variety of carbon alternatives explored in the last two decades, GNRs have certainly acquired a lot of interests. Amongst the perks, chemical synthesis of GNRs is now cost effective, they have comparable electrical performance (small effective mass, high charge mobility, and high saturation velocity) compared with the CNTs, without the need to separate metallic from semiconducting GNRs, and they can be easily tailored through chemical synthesis [58, 59].

As in the case of CNTs, confining large-area graphene carries into narrow stripes leads to a quasi-1D system exhibiting a tunable bandgap determined not only by the ribbon width w_{GNR}, but also by the edge structure [60, 61]. Logic applications, as mentioned in Sec. 1.2, require two well-defined voltage states, which in turn they depend on the current flowing through the semiconducting channel; by considering E_g within the 300 – 500 meV range, the figure (place the bandgap figure here) gives an indication circa suitable ribbon widths, which would lie within the sub-5 nm regime [62].

band structure

Historically, a first classification of graphene nanoribbons was derived by grouping the edge structures into Armchair-GNRs (AGNRs) and Zigzag-GNRs (ZGNRs). Following the success of chemical routes in producing GNRS, various kind of edges have come to light, e.g., cove, fjord, chiral, and all these can be regarded as combination of AGNRs and ZGNRs Figure.

Initial bandgap estimations through DFT calculations reported values between 2.5 and 0.5 eV. The inclusion of electronic correlation, however, becomes crucial for ultra-narrov (< 5 nm) ribbon [60, 63] and more accurate estimates derived by the group of Steven Louie based on the GW approximation reported a wider range of energy gaps from 5.5 to 1 eV.

These calculations have shown that, as rule of thumb, AGNRs can be both semiconducting and metallic, while ZGNRs should be regarded as quasi-metallic, with energy gaps lying within 0.1 - 0.7 eV. Indeed, for this kind of ribbon the application

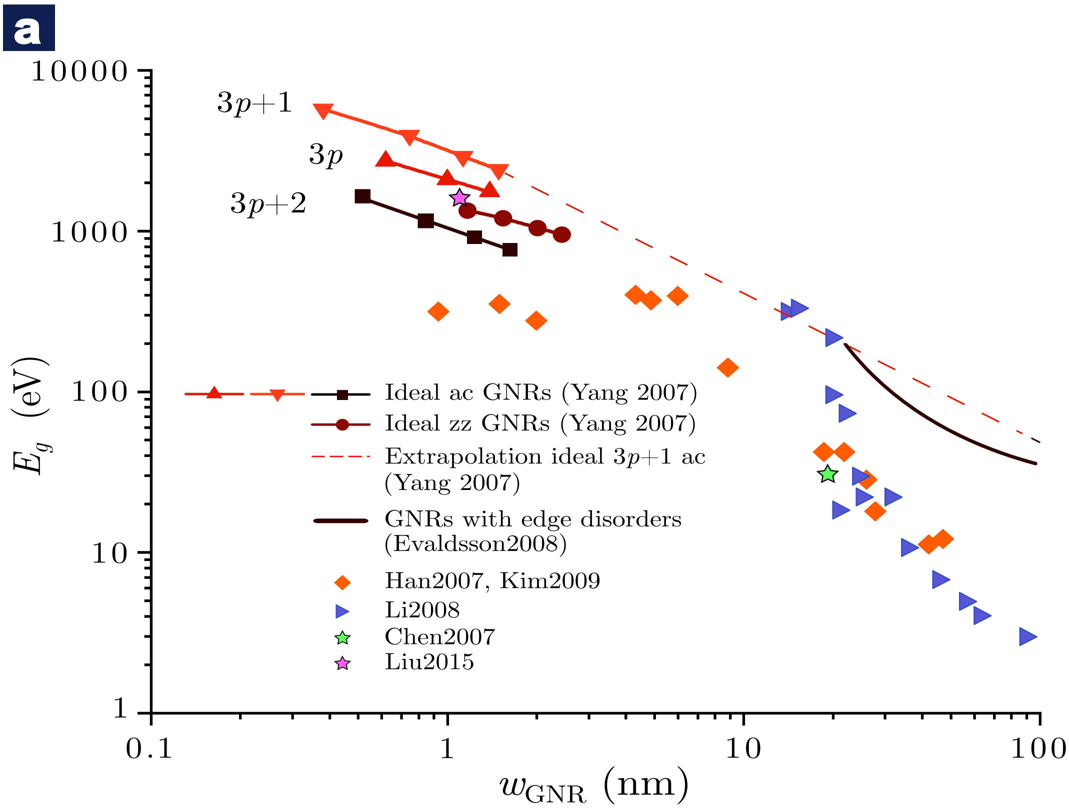

Figure 1.2: Band gap versus nanoribbon width from experiments [61, 64–66] and theoretical calculations [67, 68]. The following abbreviation are used: ac for *armchair*, zz for *zigzag*. Figure adapted and extended from [27].

of an electric field in the transverse direction of induces a spin-polarisation of the edge states is believed to be responsible for semiconducting properties of ZGNRs The simplest model for describing the energy levels for a graphene nanoribbon is the particle in the box model, where the energy should scale quadratically $E_g \propto 1/L^2$ with the ribbon extension. Experimentally, however, the band gap for AGNRs scales with the inverse of the width $E_g \propto 1/W$ [60]: by considering all the interatomic carbon-carbon distances the same, it is possible to work out an averaged formula which reports the band gap varying as $E_g = 0.8/W$, [69]. More accurate simulations have shown, however, that peripheral carbon—carbon bond lengths usually tend to be larger compared to the ribbon core ones [70]. It is predicted that the band gap will be similar to Ge or InN when the width is reduced to 2 and 3 nm, and will be similar to Si or GaAs when it is further reduced to 1-2 nm [71].

AGNR

For AGNRs, the bandgap varies significantly for different m values. AGNRs exhibit metallic behavior when $m = 3p + 2$, where p is an integer number, and they become semiconductors when $m = 3p$ or $3p + 1$. These relations have been obtained through in silico experiments, both using the tight-biding method and DFT . Figure 3 presents the isosurface plots of Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied Molecular Orbital (LUMO) of AGNRs with m=7,8,9 respectively. Interestingly, the HOMO and LUMO orbitals are localized at the terminal of the ribbon along the length direction when m=7(3p+1).

ZGNR

The band structure of ZGNRs is more interesting. Tight binding estimations have shown that if spin DOFs are excluded, the conduction band and valence band touches at the fermi level, forming a flat band. This flat band leads to a large localization of the density of states at the edges of ZGNRs, the so called *edge states*, which displays an exponential decay $\rho(\epsilon \sim \epsilon^{-\mathbf{r}})$, with an exponent depending of the distance from centre of the ribbon; the weight of these states become negligible when the width in is the order of µm. The formation of band gap is observed when the spin is considered. In particular, the ground state of ZGNRs has parallel (ferromagnetic, ↑↑, FM) spin alignment along one edge, whilst two edges have antiparallel (antiferromagnetic, ↓↓, AFM) configuration. Given that the AFM state is semiconducting while the FM is metallic, it would theoretically be possible to change the metallic state of a zigzag ribbon by applying a magnetic field in the direction transverse to that of the ribbon's extension.

The scientific interest for graphene nanoribbons started in the physics community. Although this may appear irrelevant at the first glance, we will see that this actually had an impact over the last twenty years of progress in the field. The physics community was looking for a way to open a band gap in graphene. As mentioned in Sec. 1.1, large-area graphene is a semimetal, therefore unsuitable for CMOS logic and problematic for RF.

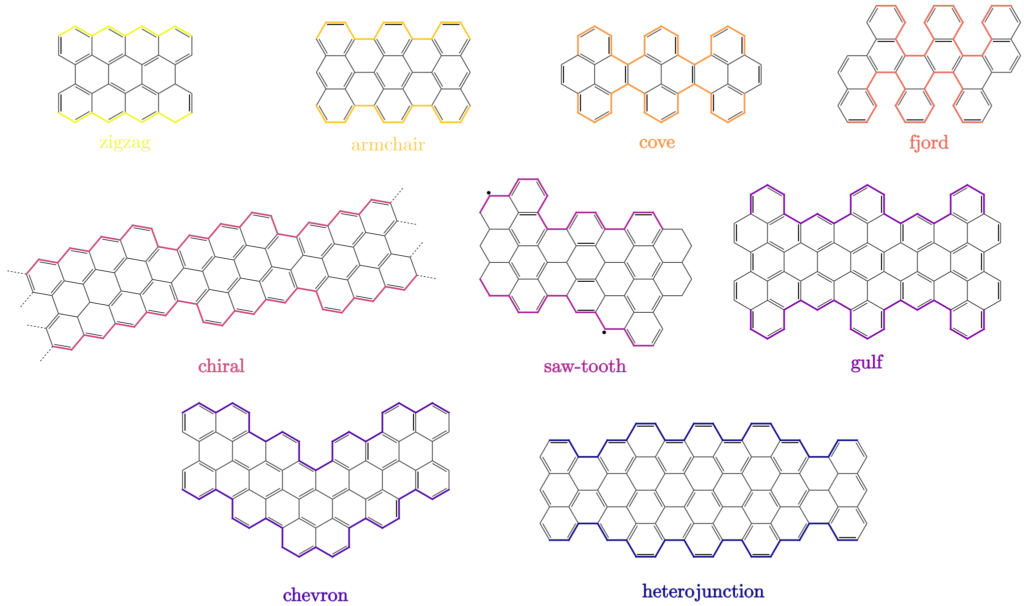

Figure 1.3: Different geometries and edge structures of graphene nanoribbons.

The first class of methods for synthesising graphene nanoribbons are methods that involve cutting or producing rectangular shapes from the graphene lattice through macroscopic methods. For this reason, they are also known as *top-down* approaches. Although they have seen an improvements in recent years, common drawbacks for this class of methods are a random distribution in the widths, a non-atomical control of the edge structures, and non-negligible impurity concentrations left after the process.

Kim and co-workers synthesized GNRs by employing the electron-beam lithography and etching technique [72]. The widths of GNRs obtained by this technique were scattered over a broad range, and most of the edges were extremely rough. An 8 nm GNR FET showed a dramatically improved on/off ratio of about 160.

Nanocutting of graphene is another promising method of GNR synthesis. In 2008, Strachan *et al.*. used thermally activated Fe nanoparticles as chemical scissors to cut graphene sheet along a specific crystallographic direction [72, 73]. Later, Ni and other catalysts were employed as well The carbon-carbon bond breaking is driven by the combustion reaction $\mathrm{Ni} + \mathrm{C_{graphene}} + 2\,\mathrm{H_2}\uparrow \longrightarrow \mathrm{Ni/Fe} + \mathrm{CH_4}\uparrow$

catalysed by the nanoparticles, thus producing hydrocarbon compounds. One of the reason that have discouraged these synthetic way is the random width distribution obtained: this is likely to be a consequence of the change in direction when nanoparticles encounter defects or edges.

edge states

Band gap opening and magnetic ordering are two main features of ZGNRs when the electron-electron correlations and the spin DOFs are considered. Steven Louie's group extended the computation also to chiral edges, elucidating the inherent role of spin-polarized magnetic edge states in these kind of ribbons. [74]. They employed a mean-field Hubbard model to calculate the band structure, assuming the magnitude for the on-site Coulomb repulsion being comparable to the hopping integral $U/t \sim 1$. The Hubbard hamiltonian using the second quantisation formalism resembles Eq. 1.11 and can be expressed as:

$$\begin{aligned}\mathcal{H}_{\text{MF}} = &- t \sum_{\langle ij \rangle,\ \sigma} \left(a^{\dagger}_{\sigma,i} a_{\sigma,j} + \text{H.c.} \right) \\ &+ U \sum_{i} \left(n_{i\uparrow} \langle n_{i\downarrow} \rangle + \langle n_{i\uparrow} \rangle n_{i\downarrow} - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle \right),\end{aligned} \tag{1.25}$$

except for the second term, which is given by the summation of spin-polarised density $n_{i\sigma} = a^{\dagger}_{i\sigma} a_{i\sigma}$ and their expectation values. The hamiltonian in Eq. 1.25 respects the electron-hole symmetry, meaning that the density of states is symmetric respect to the E_F. The solution of the model displays interesting features: two pairs of peaks split (Δ_0 and Δ_1) in the density of states are observed, one associated with the electronic band gap, and one associated with the maximum energy splitting.

The authors claim that the first peak split arises due to an AFM correlation between opposite edges, while Δ_1 is a consequence of a ferromagnetic order due to strong localised state along a single edge. Within this description, it is legit to link the behaviour of chiral GNRs with the one of spin-polarised ZGNRs; thus the total energy difference between ferromagnetic and antiferromagnetic couplings should be in the order of tens of meV per edge atom and should decrease as the width

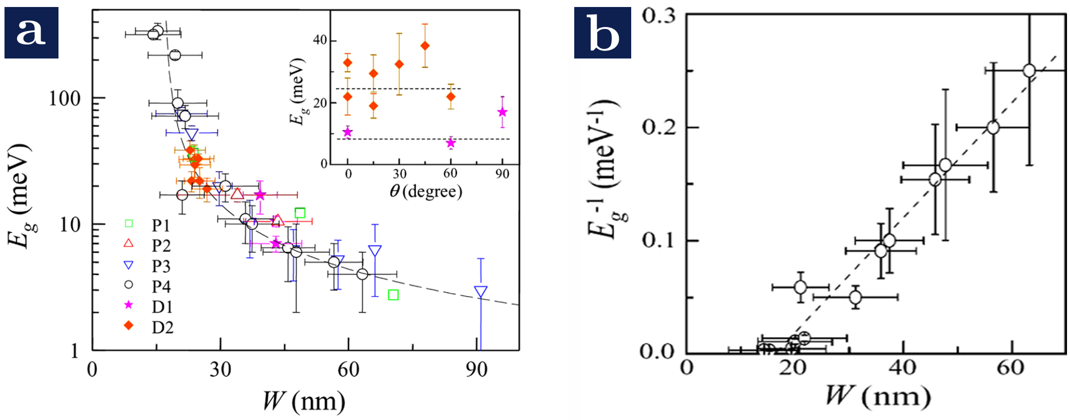

Figure 1.4: **(a)** E_g vs W for four (P1–P4) of the parallel devices and two (D1, D2) with varying orientation. The inset shows E_g vs relative angle θ for the device D1 and D2. **(b)** E_g^{-1} vs W. **(c)** Conductance vs width of parallel GNRs measured at a gate voltage V - V_g = 50 V. Squares are values measure at 300 K whereas triangle corresponds to 1.6 K. The insets show the conductivity (upper) and the inactive GNR width (lower) obtained from the slope and x intercept of the linear fit at varying temperatures. Adapted from [64].

of the ribbon increases, eventually becoming negligible when the ribbon width is significantly larger than the decay length of the spin polarized [75]. Both features are in fact recovered from the solution of Eq. 1.25.

GNRFET

The dispersion relation E($\mathbf{k}$) for a sub-10-nm-wide semiconducting GNR can be legitemately expressed by imposing a lateral confinement of the wave vectors in one direction [76]:

$$E(\mathbf{k}) = \frac{3ta}{s}|\mathbf{k}| = \frac{3ta}{2}\sqrt{k_{\parallel}^2 + k_{\perp}^2}) \tag{1.26}$$

where t is the nearest neighbour hopping parameter, a is the C-C bonding length, where $k_{\parallel}$ and $k_{\perp}$ are the components of the wave vector $\mathbf{k}$ along the transport and the confinement directions, respectively. The resulting carrier kinetic energy in case of no mid-gap states is thus $E_k = E(k) - E_g$, with $E_g = 3ta|k_\perp|/2$.

In order to quantify the mean free path for the carriers, Wang et al. [76] considered the total mean free path given from the contribution of edge, defect, and (flexural) phonons:

$$\frac{1}{\ell} = \frac{1}{\ell_{\text{edge}}} + \frac{1}{\ell_{\text{phonon}}} + \frac{1}{\ell_{\text{defect}}} \tag{1.27}$$

Without loss in generality, a simple estimation of the probability a backscattering event due to the edges should lower the mean free path as [76]:

$$\ell_{\text{edge}} = \frac{k_\parallel w_{\text{gnr}}}{P k_\perp} \tag{1.28}$$

For ribbons with widths below 20 nm and exluding interaction effects, the energy gap should scale inversely with the ribbon width [77], namely:

$$E_g \approx \hbar v_F / w_{\text{gnr}} \tag{1.29}$$

and by plugging this and the dispersion relation E_k into Eq. 1.28, it is possible to work out a dependence of the edge mean free path with the total ribbon width:

$$\ell_{\text{edge}} = \frac{w}{P}\sqrt{\left(1 + \frac{E_k}{E_g}\right)^2 - 1} \tag{1.30}$$

In one of the first graphene nanoribbon field-effect transistors (GNRFETs) prototype realised with ultra-narrow ribbons as channel material, the Dai's group reported impressive performance such as $I_{\text{on}} \approx 2000\ \mu\text{A}\,\mu\text{m}^{-1}$, $I_{\text{on}}/I_{\text{off}} \approx 10^6$, $SS = 210$ mV/dec, and $g_{\text{m}} = 900\ \mu\text{S}\,\mu\text{m}^{-1}$. Those GNRs were synthesised using sonochemistry methods, achieving widths down to $w_{\text{GNR}} \approx 2.0(5)$ nm and lengths of about $l_{\text{GNR}} \approx 236$.

using sonochemistry methods and global Si/SiO_2 backgate. In particular, nm, they were able to obtained impressive transistor performance such as . This pioneer work sparked an exciting GNRFETs research (see table place table GNRFET).

	Structural Parameter			Device Parameter	
Fabrication Approach	Edge Structure	Width (nm)	E_g (meV)	$I_{\text{on}}/I_{\text{off}}$	Mobility ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$)
Top-Down	—	5	145	47; 10^5 (5.4 K)[78]	—
	—	5	489.4 ± 19.0	$> 10^4$ (290 K)[79]	—
	—	15	120 ± 23	—[79]	—
	Mixed	10	140	10^5 (4 K)[80]	—
	—	12	100	10; 10^6 (4 K)[81]	—
	—	sub-10	200	—[82‡]	—
	—	10 – 20	10 – 15	4.5; 10 (20 K)[83]	—
	Closed	2.8	494	10^5	2443_{μ_h}
	Closed[84]	9.9	$891/[\text{w}_{\text{gnr}} - 1.0]$	$< 10^2$	3776_{μ_h}
	—	100 – 400	18 – 1	20 – 3.5[85]	—
Bottom-up	Cove	0.69 – 1.13	1880	$>^{\dagger} 10^{10}$	
	AGNR	0.95	2100	$^{\dagger}10^5$	
	Chevron	1.73	3100 ± 400	—	
	—	2	—	$> 10^6$	
	AGNR	0.63	1690 ± 100	$>^{\dagger} 10^10$	
	AGNR	0.95	1690 ± 70	—	
	AGNR	1.27	1130 ± 50	$^{\dagger}10^9$	
	AGNR	1.37	1600 ± 400	—	
	AGNR	1.58	1030 ± 40	$^{\dagger}10^8$	
	AGNR	5 – 9	≤ 400	$10^5 - -10^6$	
	ZGNR	6	1400	—	
Epitaxial	—	900	—	—	
	AGNR	2 – 40	> 500	—	
	—	40	—	10; > 25 (4 K)	

Table 1.4: Comparison amongst different FET realised with graphene nanoribbon as conducting channel. Top and bottom.

While it has been advanced in the past the possibility to obtain graphene nanoribbons from nanotubes cite work on unzipped CNT, it is relatively recent the proposal of using high-pressure conditions and thermal treatments to produce ultranarrow (sub-10-nm-wide) GNRs from CNTs, resulting in average lengths in the order of μm range.

The p-type on-state conductivity of the ribbon channel ($\mathrm{w_{gnr}} \approx 2.8$ nm) when placed in a MOSFET geometry is:

$$\sigma = G_{\mathrm{sd}} L_{\mathrm{ch}} / w_{\mathrm{gnr}} = 7.42 \text{ mS} \tag{1.31}$$

and $I_{\mathrm{on}}/I_{\mathrm{off}} \approx 10^5$. Using Eq. 1.19, the authors reports a $E_g \approx 494$ meV. A further comparison was carried out by comparing the mobilities values obtained for while increasing up to 10 nm the GNRs width

Hole field-effect mobility was calculated using the classic MOSFET model through the relation:

$$\mu = g_{\mathrm{m}} L / (C_{\mathrm{gs}} V_{\mathrm{sd}}) \tag{1.32}$$

and estimating the gate capacitance with an electrostatic 3D model.

Interesting enough, the hole mobilities values (2,443 $\mathrm{cm^2V^{-1}s^{-1}}$ for the 2.8 wide and 3.776 $\mathrm{cm^2V^{-1}s^{-1}}$ for the 10 nm wide) are amongst the best values obtained so far. The role of strained bulbs and non-spherical cross section at the two edges is not clarified yet and thus could reproduce unpredictable effects due to uncontrolled scattering mechanism.

2
Foundations

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The lateral confinement of the carriers makes GNRs a quasi-1D system. The *quasi* referes to the fact that the electron is confined across two directions. This fact became true not just until recently with the advent of bottom-up techiniques, which are capable synthesising atomically-precise GNR with width below 5 nm.

In general, quantum transport phenomena takes place on a mesoscopic scale, that is, when dealing with condensed phase systems, in particular (but not exclusively) those of micro-nano dimensions, for which behaviours not present in macroscopic bodies are observed. The different behaviour of mesoscopic phases compared to macroscopic ones liesin the wavelength nature of electrons. When the size of the sample L (which can be different for the three directions in space, L_x, L_y, L_z) becomes small compared with the characteristic lengths for the motion of electrons, then the conductivity depends on the size L.

There are three kinds of relevant characteristic lengths: the Fermi wavelength $\lambda_F = h/\sqrt{2m_e E_F} = 2\pi/k_F$, and the coherence length L_ϕ, which is the spatial distance that an electron travels before the phase of its wave function changes of 2π.

Additionally, two characteristic lengths associated with the mean distance traveled after elastic (l_{el}) and inelastic (l_{in}) collisions are also introduced; for inelastic collision, it is convenient to consider the mean distance before losing an amount of energy $k_B T$.

For a metal with carrier concentrations $n \sim 10^{23}$ cm^{-3}, $l = 1 - 10$ nm, $\lambda_F = 0.5$ nm, and $L_\phi = 0.5$ µm, where l is the mean free path and includes both elastic and inelastic events. In comparison, for carrier concentrations $n \sim 10^{11}$ cm^{-2}, graphene displays $l = 50$ nm$-3^\dagger$ µm, $\lambda_F = 2\sqrt{\pi/n}$ nm, and $L_\phi = 0.5$ µm [87], while MW-CNTs $n \sim 10^{11}$ cm^{-2}, $l = 1$ µm, $\lambda_F = 0.74$ nm, and $L_\phi = 100$ nm [88].

2.1 Charge Transport In Zero-Dimensional Objects

Quantum dots (QDs) defined through lithography can be seen as a metallic island filled with electrons, coupled to one or multiple source, drain and gate electrodes (Figure). While source and drain are coupled both capacitively thorugh the island thanks to tunnel junctions, the gate (or multiple gates) is just capacitatively coupled.

One way to describe this system is through the constant interaction model, which allows to derive important electrical transport information within the *orthodox* Coulomb framework. The assumptions made within this model are the following: all Coulomb interactions, both between electrons on the dot and between the dot

†For suspended graphene, $l(4\text{ K}) \approx 100$ nm, while Mayorov et al. $l \approx 3$ µm at 290 K for graphene sandwiched between two hBN crystals [86].

and the leads, are described by a single constant capacitance C given by the total sum of the single individual capacitance between the dot and the terminals:

$$C_{\text{tot}} = \sum_{i=1}^{\text{terminals}} C_i, \tag{2.1}$$

where i runs over all the terminals coupled to the central island. The second important assumption is that the total hamiltonian can be modelled as a sum of single-particle hamiltonians; thus, the total energy can be expressed as a sum of single-particle energies E_i corresponding to individual, non-interacting energy levels.

Under these assumptions, the total dot energy given for a quantum dot in a three-terminal device with N electrons becomes:

$$U(N) = \sum_{i}^{N} E_i + \frac{[e(N - N_0) + C_{\text{S}}V_{\text{S}} + C_{\text{D}}V_{\text{D}} + C_{\text{G}}V_{\text{G}}]^2}{2C_{\text{tot}}}, \tag{2.2}$$

where eN_0 is the charge present on the dot at zero bias and zero gate voltage due to positive background charges in the substrate. Central quantity to evaluate whether the current can flow through the dot is its the chemical potential, which is the minimum energy required to add an extra electron to the dot, and can be expressed as:

$$\begin{aligned}\mu_{\text{dot}}(N) &= E_{\text{dot}}(N) - E_{\text{dot}}(N-1), \\ &= \left(N - N_0 - \frac{1}{2}\right) E_{\text{C}} - \frac{E_{\text{C}}}{e}\left(C_{\text{S}}V_{\text{S}} + C_{\text{D}}V_{\text{D}} + C_{\text{G}}V_{\text{G}}\right) + E_N,\end{aligned} \tag{2.3}$$

where $E_{\text{C}} = e^2/C_{\text{tot}}$ is the charging energy.

2.2 Molecule–Lead Couplings

2.2.1 Weak Limit

The weak coupling regime is defined in the limit where $\Gamma \ll E_C, E_{\text{add}}(N), k_BT$, and is dominated by Coulomb blockade. The effect shows most clearly in a charge stability diagram, which is obtained by measuring the gate and bias dependence of the current and then plotting it in a two-dimensional map Figure. The Coulomb blockade manifests as diamond shaped regions of very low current, so called Coulomb diamonds, in which transport is blocked. In these regions the energy cost to add an electron onto the QD exceeds the available energy and therefore no current can flow through the QD Figure.

The blockade areas are separated by regions of high current, the Sequential Electron Tunnelling (SET) zones, where current flow through the QD is allowed via sequential electron tunnelling. Each Coulomb diamond corresponds to a certain number of electrons N occupying the QD where the number of electrons changes sequentially between neighbouring diamonds. When looking closely at the SET regions, often lines running parallel to the edges of the Coulomb diamonds can be observed. These originate from orbital excited states, spin states or vibrational states, which will be discussed later. These states are able to contribute to the current flow through the QD if their energy levels are reached by applying a higher bias voltage Figure. Such excited or vibrational states can yield information about the QD's energy spectrum and are especially helpful when attempting to identify specific molecules as each molecule has a distinctive vibrational fingerprint.

Furthermore, the height of the Coulomb diamond in the stability diagram is equal to the addition energy of the QD or molecule Figure. While it is possible to readily read out the addition energy and the energy of excited states, it is very challenging or almost impossible to infer the charge state of the QD, that is the number of electrons on the dot, from the stability diagram. Since the QD is part of the device and therefore a larger system, the neutral charge state can be deceiving when only considering the dot itself. Assuming the QD is at no applied bias and no gate voltage and therefore seemingly in the neutral charge state, it might still be partially charged due to interactions with the junction or background charges in the substrate (cf. Eq. 2.2).

One can correlate the stability diagram or a gate sweep measuring the conductance or current through the quantum dot to the different level alignments of the contacts and the dot orbitals Figure. When the electrochemical potential of the source and drain is aligned with an energy level of the dot, transport through it is allowed and results in a current peak in the gate trace. The corresponding transport mechanism is called resonant tunnelling. While the levels are aligned the number of electrons on the dot fluctuates between N or $N+1$ as electrons hop on and off the dot. In between the peaks the electrochemical potentials are not aligned, so there is no resonant tunnelling due to Coulomb blockade, and the number of electrons on the dot is fixed. Additionally, various capacitances between the QD and the contacts as well as the gate lever arm (or gate coupling parameter) can be extracted from the slopes of the diamonds as the negative slope and positive slope are equal to $-|e|C_\mathrm{G}/(C - C_\mathrm{S})$ and $|e|C_\mathrm{G}/C_\mathrm{S}$, respectively. Since the gate lever arm is given by $\alpha = C_\mathrm{G}/C$ one can thereby quantify the strength of the gate coupling.

2.2.2 Intermediate Limit

Lastly, the intermediate coupling regime has attributes of both limits, it still exhibits a noticeable Coulomb blockade, but also shows transport within the zero current region. The two most noticeable new transport channels available in this regime are elastic and inelastic cotunnelling: these higher-order transport mechanisms involve the simultaneous tunnelling of two electrons (Figure). Initially, the QD is in a state with an unoccupied excited state still below the electrochemical potential of the leads, in contrast to resonant tunnelling where the excited state is between the source and drain electrochemical potential. When the electron tunnels out of the QD onto the drain, the QD is left in a virtual state and there are two possibilities for the orbit to be refilled. First, the new electron simultaneously tunnelling in from the source lead occupies the ground state again, therefore leaving the QD in exactly the same configuration as before. This is called elastic cotunnelling (maybe Figure?). If the new electron, however, tunnels into the excited state, the QD has changed its energy and the process is called inelastic cotunnelling (Figure?). The electron can subsequently relax into the ground state, rendering the QD again in the initial configuration. Importantly, for inelastic cotunnelling to take place, a bias larger than the excitation energy of the dot has to be applied and since the QD excitation is created at the expense of the bias applied, electron energy is conserved. Overall, both elastic and inelastic cotunnelling lead to conduction in the blocked Coulomb diamond region.

2.2.3 Strong Limit

In the strong coupling limit $\Gamma \gg E_C, E_{\text{add}}(N), k_B T$ In this regime quantum dot and lead's states hybridise, the Coulomb blockade is smeared out by quantum fluctuations of the electric charge, and the main transport mechanism is now played by elastic coherent tunnelling. In this limit, the conductance can be approximated as $G \approx \frac{2e^2}{h}\mathcal{T}(E)$, where $\mathcal{T}(E)$ is the transmission probability for all the conduction channels thorugh the quantum dot evaluated at the Fermi energy. In the case of a single, well-isolated energy level, it is possible to recover the Breit–Wigner model for a single resonance.

2.2.4 Landauer Theory

In order to make predictions about the electronic and thermoelectric properties of QDs, a good model for electronic transport in mesoscopic to molecular-scale devices is necessary. In the limit of predominantly SET and assuming coherent transport, extending the Landauer formalism (originally a model for electrical conduction) towards thermoelectric effects has proven successful and is widely used. The Landauer formalism is a concept formulated in 1947 by Rolf Landauer that models conduction as transmission. The basic assumption is that two electron reservoirs are connected by a scatterer (QD island for the transport measurements presented here) through which electron transport is possible. The transport is then determined by the probability of electrons moving from one reservoir to the other through the scatterer. This probability is always between 0 and 1, which means there is an upper limit on conductance.

Employing the Landauer approach, the current through such a structure can be written as:

$$I = \frac{2e}{h} \int_{-\infty}^{+\infty} \mathcal{T}(\epsilon)[f_{\text{L}}(E) - f_{\text{R}}(E)]\text{d}\epsilon, \tag{2.4}$$

where $\mathcal{T}$ is the transmission probability of an electron with single-particle energy ϵ to move across the source and drain leads. The Fermi-Dirac statistics $f_{\mathrm{FD}}(\epsilon, T) = \frac{1}{1+e^{\epsilon_i - \mu / k_B T}}$ details the average number of fermions (electrons in our case) in a single-particle state i with energy ϵ_i in thermodynamic equilibrium at temperature T. Therefore, it is used to quantify how many electrons from the source leads have a certain energy to be able to cross the molecular junction: this event, as everything in quantum mechanics, is detailed by the amplitude of the transmission probability $\mathcal{T}(\epsilon)$.

One can calculate the number of electrons in a crystal by multiplying the density of states $N(\epsilon)$ with the FD distribution and summing up this product for all energies:

$$n = \int_0^\infty f_{\mathrm{FD}}(\epsilon) N(\epsilon) \mathrm{d}\epsilon, \tag{2.5}$$

The density of states $N(\epsilon)$ denotes the number of electron states per unit of energy and volume and differs for different materials. In such a system the chemical potential is defined as the amount of energy needed to add an additional electron to the system and naturally this quantity has an inherent temperature dependence as the energy of the system changes with T. An approximation for the chemical potential as a function of temperature and energy has been developed using the Sommerfeld expansion (cf. Appendix A): since we are interested in low-temperature properties, we will consider only terms up to the second order of Eq. A.7:

$$\mu \approx \epsilon_F - \frac{\pi^2}{6} \left(k_B T\right)^2 \frac{1}{N\left(\epsilon_F\right)} \frac{\partial N(\epsilon)}{\partial \epsilon}\bigg|_{\epsilon=\epsilon_F} \tag{2.6}$$

As is evident from this equation, the chemical potential decreases as the temperature rises, resulting in a potential gradient whenever a temperature gradient is applied. Any difference in T will therefore lead to a flow of electrons as the cost of adding another electron to the hot side would be lower than adding it to the cold side. This results in the build-up of a thermoelectric voltage in an open circuit.

By multiplying this probability with the elementary charge e gives the current for one energy level with a factor of two added due to spin degeneracy. After adding all of these possible transport channels together, which is done by taking the integral over all energies, and accounting for dimension normalization via the Planck constant h it is possible to calculate the total current I across the junction.

The electrical conductance is then retrieved by taking the derivative of the current:

$$\begin{aligned} G &= -\frac{2e}{h}\int_{-\infty}^{+\infty} \mathcal{T}(\epsilon)\frac{\partial f(T,\epsilon)}{\partial \epsilon}\mathrm{d}\epsilon, \\ &= -G_0\int_{-\infty}^{+\infty} \mathcal{T}(\epsilon)\frac{\partial f(T,\epsilon)}{\partial \epsilon}\mathrm{d}\epsilon, \end{aligned} \tag{2.7}$$

As we will see in more detail later (cf. Sec. 2.5), it is possible to compute the set of real eigenvalues for the transmission thorugh different approaches with varying degrees of complexity. One of them, which it has been used in cf. Ch. 5, calcultes the product between the tunnel couplings and Green function terms. Other and more sophisticated approaches can be found in literature [89].

2.3 Thermoelectric Phenomena in Zero-Dimensional Structures

In this section we will focus on those electrical phenomena arising when a thermal gradient is present between two junctions [90]. As it is well know, macroscopically, the thermoelectric power that arises when two different materials are welded together in two points (junctions) is proportional to the temperature drop establishing when one of the junction is heated up at temperature $T_1 > T_2$; i.e.:

$$\Delta V = S\Delta T \tag{2.8}$$

where ΔV denotes the electrical potential drop and S is the proportionality factor called Seebeck coefficient. For reasons that will become clear later, it is prefferred to define the as the negative voltage difference ΔV induced per unit temperature change ΔT, namely $S = -\lim_{\Delta T\to 0}\frac{\Delta V}{\Delta T}$.

There are two approaches used to model the electronic distribution in a solid when external paramaters such as heat and/or non-uniform fields are applied. The

Boltzmann approach infers the dynamics of a diluted gas of semi-classical particles: in the semiclassical description of transport, the electron ensamble is described by a probability distribution function $f(\vec{r}, \vec{k}, t)$, which details the number of electrons that a certain time t are within a volume $\mathrm{d}\vec{r}$ with wave vector within the interval $\mathrm{d}\vec{k}$. In the case where quantum effects play a role in transport, that is if the potential of the structure varies sharply with respect to the particle average thermal wavelength, the Boltzmann approach is no longer valid.

The second applicable framework is the Landauer theory (cf. Sec. 2.2.4) which is used for low temperature thermoelectric calculations in mesoscopic structures cite Cuevas. The Landauer formalism has the advantage of assuming a constant mean-free path, rather than a constant relaxation time of the electrons as the Boltzmann approach does, which is more useful in experiments. In addition, the Landauer approach is valid in the ballistic and quasi-ballistic regime.

It is possible to formulate a similar expression for the flow of heat Q from one reservoir to the other by multiplying the particle transmission with the particle energy $(\epsilon - \mu)$:

$$I = \frac{2}{h} \int_{-\infty}^{+\infty} (\epsilon - \mu)\mathcal{T}(\epsilon) f(\epsilon, T))\mathrm{d}\epsilon. \tag{2.9}$$

If the electrical or thermal gradient are treated within a linear approximation, such that $\Delta T = T_0 - T$ and $e\Delta V = \mu_0 - \mu$, the Fermi-Dirac distrubtion can be rewritten in a Taylor series and stopping and the first order of the expansion:

$$f(T, \epsilon) = f^0 - \frac{\partial f}{\partial \epsilon}\left[e\Delta V - \frac{\epsilon - \mu}{k_B T}\Delta T\right] \tag{2.10}$$

Thus, the total amount of current flowing within the device is given from the thermoelectric current due to the thermal gradient ΔT applied to the junction and the electric current due to bias applied between source and drain:

$$I = G\Delta V - L\Delta T, \tag{2.11}$$

$$Q = LT\Delta V - K\Delta T. \tag{2.12}$$

Here G, L, and K denotes the electric, the thermoelectric and the thermoconductance, resepctively. The relation expressed in Eq. 2.11 is particularly convenient, as through the Wiedemann-Franz law it is possible to retrieve the Seebeck coefficient. For an open circuit we will then have:

$$I = 0 = G\Delta V - L\Delta T,$$

$$\sigma\Delta V = \alpha\Delta T,$$
$$\frac{\Delta V}{\Delta T} = \frac{\alpha}{\sigma} \tag{2.13}$$

where we called with σ the conductivity. This relation allows us to identify the Seebeck coefficient as:

$$S = -\frac{\Delta V}{\Delta T} = \frac{\sigma}{\alpha}. \tag{2.14}$$

2.4 Density Functional Theory

What are the relevant parameters/quantities that we are interested to calculate ab initio Why we opted for a pseudopotential such as SIESTA rather than others (quantum espresso). Explain why calculating the isolated gas phase all-electron electronic structure is legit for a qualitative see how the electron density is distributed over the system.

Remember also to emphasise the difference from stricly and non-stricly localised functions used for computing the electronic properties: when the functions are stricly localised they are zero outside a given radius from the atom they are associated with and so the radius is a key quality determined parameter for the basis set

2.5 Non-Equilibrium Green Function Theory

This section is needed to explain formally the results from transiesta and TB-trans calculations

2.6 Room Temperature DC Measurements

Describe here the probe station setup

2.7 Dipstick DC Measurements

Describe here the dipstick used for both liquid N_2 and liquid He

2.8 Cryogenic DC Measurements

Describe here the delft box. Describe how the delft box modular structure enables to adapt two and multi-terminal DC measurements

2.9 Cryogenic AC Measurements

Describe here the delft box + lock-in setup and why it is so complicated to set it usepackage

2.10 Thermoelectric Measurements

Describe here the setup used for the thermoelectric devices and why it needs a dedicated section Describe also here from a setup point of view which complications have raised during the setup (direct lines instead of dc lines, the need to use nano-D connectors and to design the a new pcb board).

3
Fabrication

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3.4 Deposition Process and Related Phenomna 45

3.1 General Fabrication Procedure

The steps are well known: chip preparation, e-beam, development, metal deposition and lift-off, graphene transfer old process, graphenea graphene

3.1.1 Graphene Preparation and Wet Transferring

Initially the graphene was transferred on top of the substrate through the wet transfer method. The graphene was supplied by Graphene ain 5 x 5 cm2 rectangles, with a PMMA front-layer and a back-layer of partially removed graphene. Before transfer, the back-layer had to be fully removed via etching, and the remaining stack appropriately sized for the substrates for this work. The graphene arrived in a vacuum-sealed pack, which had to be transported to the cleanroom before being opened and the following preparation steps initiated. Figure 6.2 depicts the main stages involved in preparing the graphene for transfe

3.2 Graphene Nanogaps

Here three methods have been exploited: standard feedback electroburning, gate-dependent electroburning, and litographic patterning of graphene nanogap

3.3 Chip Design

Here the designs for the different chips that have been used. Give a proper credit to Dr. Alex Gee for the fabrication of thermoelectric and the common gate ones.

3.3.1 2T Design

How convenient and 'quick' is the usage of a backgate device. List, as usual, pro and cons

3.3.2 3T Design

Why the need to use 3T over 2T

Local Gates from Waterloo

Give proper credit

Interconnected Drain and Gate

The one I made and also made with the help of Alex

3.3.3 Thermoelectric Device

Common and different features of the thermoelectric

3.4 Deposition Process and Related Phenomna

Why it has been important to try different deposition techiniques How this has been qualitatevely addressed.

4

Room Temperature Graphene Nanoribbon Single-Electron Transistor

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4.4 Summary 48

4.1 Introduction

A brief intro why single-electron transistors with gnr are interesting A brief intro to what we are gonna do/measure in this chapter

4.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks is intrinsically scarse in molecular electronics.

4.3 Electrical Transport Measurements

What type of experiments have been performed and which tricks have been used in triton to do so. What are the evolution in temperature of the zero-bias, low-bias and high bias conductance. Look at the dependence in temperature and explain possible interpretations: why it is not a Luttinger Liquid (the gnr has a bandgap) and what possible phonon modes are responsible at some point for desctructing the

4.4 Summary

5

Chemical Doping of the Bandgap for Cove Graphene Nanoribbons

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5.1 Introduction

A brief intro on how why it is important to study this

5.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

5.3 *Ab-initio* Experiments

What DFT and NEGF have been used and for what.

5.4 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

5.5 Summary

6

Thermocurrent Spectroscopy of cove Graphene Nanoribbons

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6.4 Summary 52

6.1 Introduction

A brief intro on how why it is important to study this what has been done.

6.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

6.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

6.4 Summary

7

Transport Spectroscopy of Biaryl-linked Dy–Tb Porphyrin

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7.1 Introduction

A brief intro on how why it is important to study this

7.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

7.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

7.4 Summary

8
Conclusions and Outlook

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8.1 Conclusions

Here the final conclusions

8.2 Outlook

Appendices

A
Sommerfeld Expansion

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A.1 Low-temperature Limit 59

Let us consider the following integral:

$$I = \int_{-\infty}^{\infty} h(E) f_0(E) \mathrm{d}E, \tag{A.1}$$

where $h(E)$ is a regular function, that it is continous and differentiable at any order with respect to E, and it is null for $E \longrightarrow -\infty$, whereas:

$$f_0(E) = \frac{1}{1 + e^{E-\mu/k_B T}},$$

is the Fermi-Dirac distribution.

A.1 Low-temperature Limit

We demonstrate that, at low temperatures, it is possible to obtain an analytical expression for the integral Eq. A.1 which reads:

$$I = \int_{-\infty}^{\mu} h(E)\mathrm{d}E + \frac{\pi^2}{6} \left(k_B T\right)^2 \left[h'(E)\right]_\mu + \frac{7\pi^4}{360} \left(k_B T\right)^4 \left[h'''(E)\right]_\mu + \dots \tag{A.2}$$

For low-temperature here I mean those which satisfy the condition $T \ll T_F$, where $T_F = E_F/k_B$ is the Fermi temperature for an electron gas. Let us define:

$$g(E) = \int_{-\infty}^{E} h(E)\mathrm{d}E.$$

Through an integration per parts, Eq. A.1 becomes:

$$\begin{aligned} I &= \int_{-\infty}^{\infty} h(E) f_0(E)\mathrm{d}E = [g(E)f_0(E)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} g(E)\frac{\partial f_0(E)}{\partial E}\mathrm{d}E, \\ &= \int_{-\infty}^{\infty} g(E)\left(-\frac{\partial f_0(E)}{\partial E}\right)\mathrm{d}E, \end{aligned} \tag{A.3}$$

being the first term null as $g(E) = 0$ for $E \longrightarrow -\infty$ and $f_0 = 0$ for $E \longrightarrow \infty$. At low temperatures the function $-\frac{\partial f_0}{\partial E}$ has appreciable values only near μ. It is possible to devolop a series expansion of $g(E)$ around μ:

$$g(E) = g(\mu) + \sum_{n=1}^{\infty}\left[\frac{(E-\mu)^n}{n!}\right]\left[\frac{d^n g(E)}{dE^n}\right]_{\mu} \tag{A.4}$$

This allow us to write the integral Eq. **??** as:

$$\begin{aligned} I &= g(\mu) + \sum_{n=1}^{\infty}\int_{-\infty}^{\infty}\frac{(E-\mu)^n}{n!}\left[\frac{d^n g(E)}{dE^n}\right]_{\mu}\left(-\frac{\partial f_0(E)}{\partial E}\right)dE, \\ &= g(\mu) + \sum_{n=1}^{\infty}\int_{-\infty}^{\infty}\frac{(E-\mu)^{2n}}{(2n)!}\left[\frac{d^{2n-1} g(E)}{dE^{2n-1}}\right]_{\mu}\left(-\frac{\partial f_0(E)}{\partial E}\right)dE. \end{aligned} \tag{A.5}$$

where we killed the odd terms in the series as these are integrals of odd functions (note that $\frac{\partial f_0(E)}{\partial E}$ is an even function with respect to E)

By using the change of variable $x = \frac{E-\mu}{k_B T}$, the integral can be written as:

$$I = g(\mu) + \sum_{n=1}^{\infty} c_{2n}\left(k_B T\right)^{2n}\left[\frac{d^{2n-1}h(E)}{dE^{2n-1}}\right]_{\mu} \tag{A.6}$$

$$c_{2n} = \frac{1}{(2n)!}\int_{-\infty}^{\infty} x^{2n}\left[-\frac{d}{dx}\frac{1}{e^x+1}\right]dx, \tag{A.7}$$

Computing the coefficients is done throught the integration per parts: by using the following relation:

$$\int_0^\infty x^{n-1} \frac{1}{e^{x+1}} dx = \left(1 - 2^{1-n}\right) \Gamma(n)\zeta(n), \tag{A.8}$$

it is possible to work out a closed form for the coefficients, namely:

$$\begin{aligned} c_{2n} &= \frac{1}{(2n)!} \left[- \left[x^{2n} \frac{1}{e^x + 1} \right]_{-\infty}^{\infty} + 2n \int_{-\infty}^{\infty} x^{2n-1} \frac{1}{e^x + 1} dx \right] \\ &= \frac{1}{(2n)!} \left[0 + 4n \left(1 - 2^{1-2n}\right) \Gamma(2n)\zeta(2n) \right] \end{aligned} \tag{A.9}$$

and we obtain

$$\Gamma(n+1) = n! \quad \zeta(2) = \frac{1\pi^2}{6} \quad \zeta(4) = \frac{\pi^4}{90}$$

The first kind of intellectual and artistic personality belongs to the hedgehogs, the second to the foxes . . .

— Sir Isaiah Berlin [91]

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